

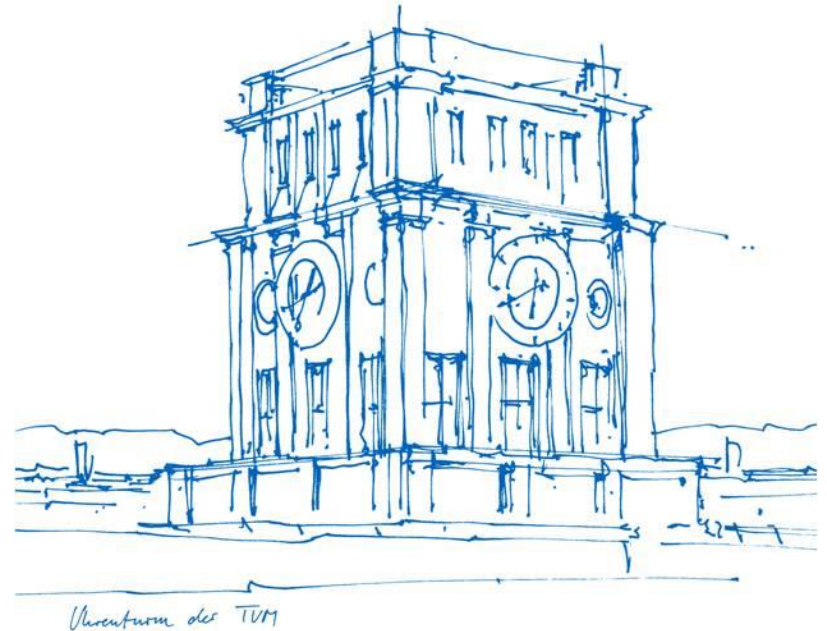
Basics of FIM: Statistical Learning

3: OLS Regressions

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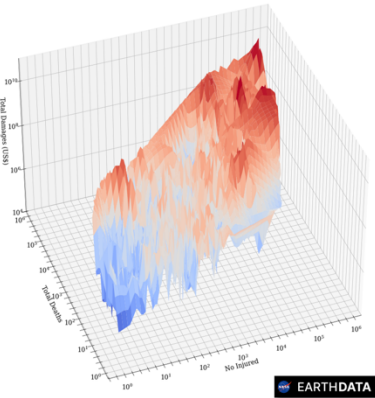
- Simple linear regression
- Multiple linear regression
- Extensions to linear regression

- Simple linear regression
 - Estimating the Coefficients
 - Assessing the Accuracy of the Coefficient Estimates
 - Assessing the Accuracy of the Model
- Multiple linear regression
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Quick rewind: what we already know about learning $f(X)$.



Why have a function from X to Y in the first place?



$$\text{Total Damage (US\$)} \approx f(\text{Homeless, Total Death, Injured})$$

$$\hat{Y} \approx \hat{f}(X)$$

- With a good $\hat{f}$ we can make predictions of Y at new points $X = x$
- We can understand which components of $X = (X_1, X_2, \dots, X_p)$ are important in explaining Y

Statistical Learning (Basics of FIM) | Prof. Dr. Niklas Kühl

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We bet the world is a line plus noise.

Linear regression is a simple approach to supervised learning. We believe about the world that: $Y \approx f(X) + \varepsilon$. In the case of linear regression, we assume that the dependence of Y on X is linear:

$$Y = \beta_0 + \beta_1 \times X + \varepsilon$$

Annotations for the equation:

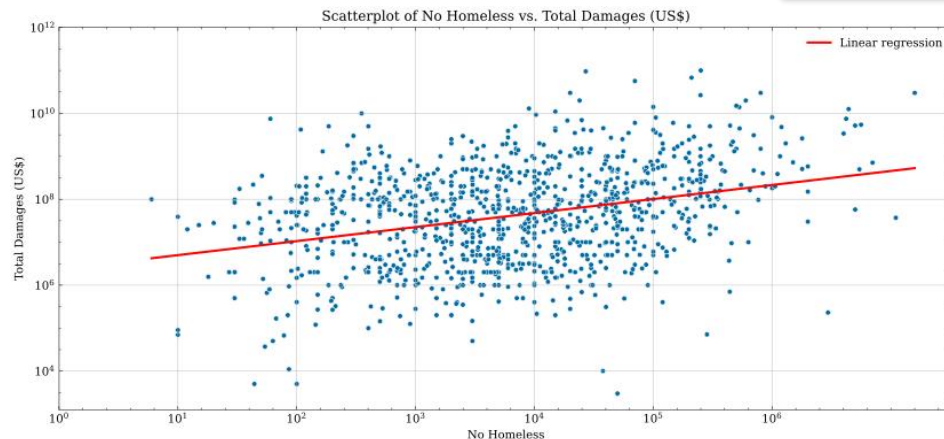
- β_0 : y-axis intercept
- β_1 : slope
- ε : error term

... but we can not observe this entity

... instead we have a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$

... from which we learn the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$



True regression functions are never linear!

Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

We pick the line that minimizes squared prediction errors.

$$Y = \beta_0 + \beta_1 \times X + \varepsilon$$

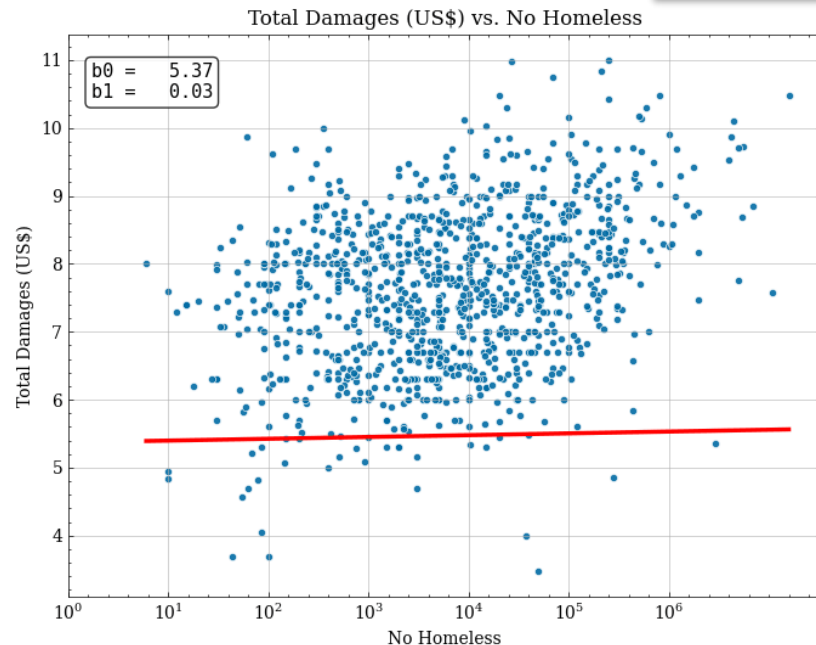
β_0 and β_1 are unknown constants that we want to estimate ($\hat{\beta}_0$ and $\hat{\beta}_1$ respectively).

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$\min \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

*... such that we
minimize the error of
our estimations $\hat{y}$*

$$= \min \sum_{i=1}^N (y_i - \hat{\beta}_0 + \hat{\beta}_1 x_i)^2$$

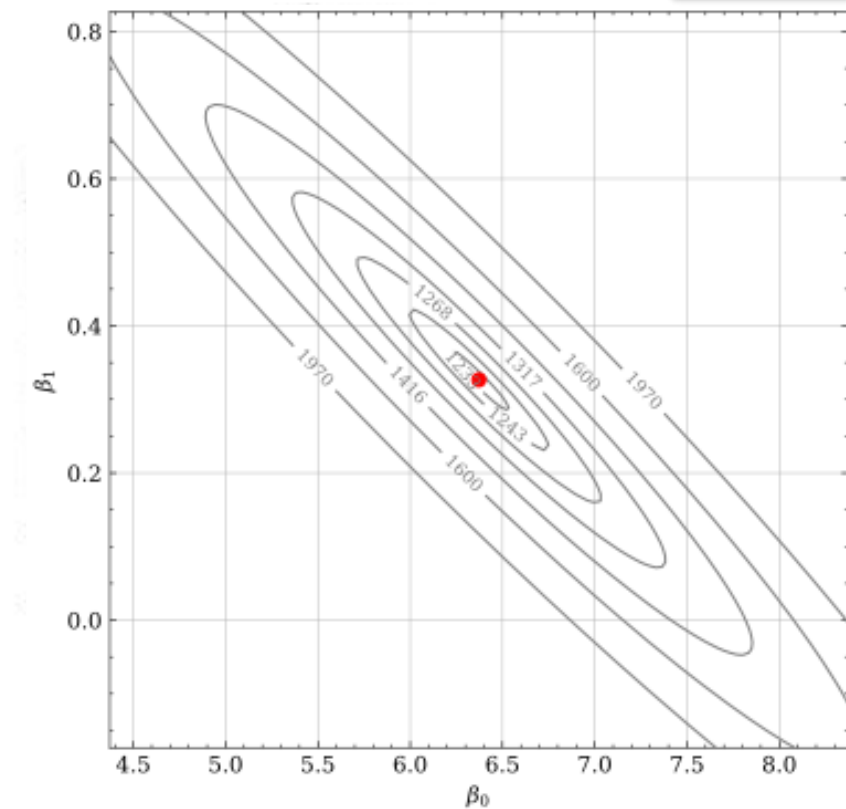


For different combinations of our parameter estimates we can compare the residual sum of squares.

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$= \min \sum_{i=1}^N (y_i - \hat{\beta}_0 + \hat{\beta}_1 x_i)^2 = \min \sum_{i=1}^N \varepsilon_i^2$$

Residual sum of squares (RSS) 



OLS has a closed form solution.

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$= \min \sum_{i=1}^N (y_i - \hat{\beta}_0 + \hat{\beta}_1 x_i)^2 = \min \sum_{i=1}^N \varepsilon_i^2$$

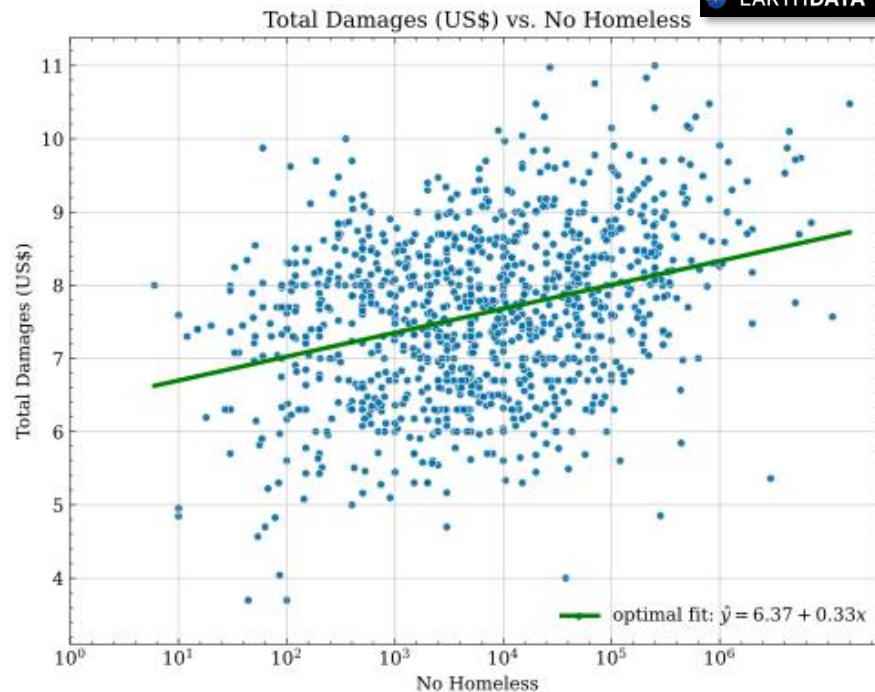
Residual sum of squares (RSS) 

Using some calculus, one can show that the minimizers are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

mean of all x  *mean of all y* 

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$



We have a closed-form solution, deterministic by construction. The optimum is computed directly, not searched for.

- Simple linear regression
 - Estimating the Coefficients
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- Extensions to linear regression

Standard errors tell us how much each coefficient would vary across samples.

How good are our estimates $\hat{\beta}_0$ and $\hat{\beta}_1$?

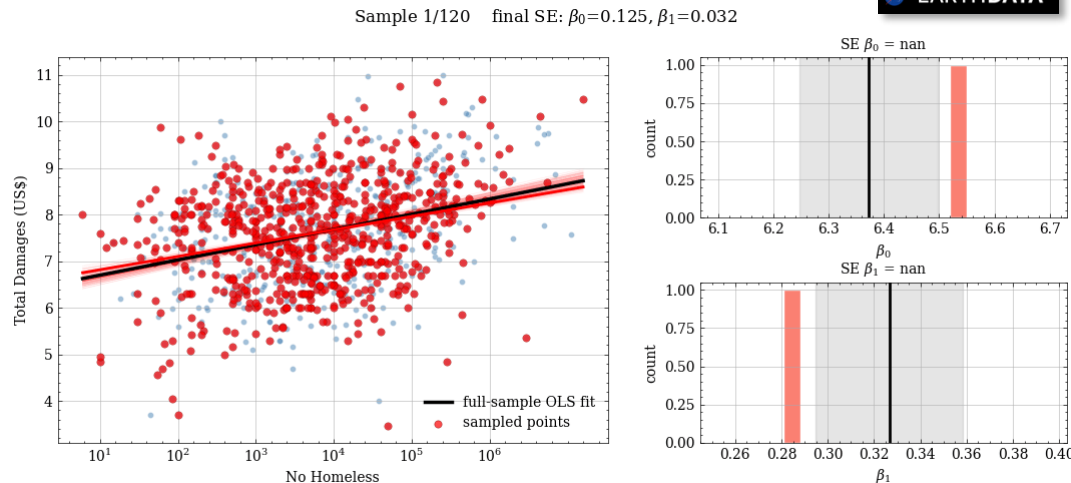
We believe about the world that: $Y \approx f(X) + \varepsilon$.

... but we can not observe this entity

... we have a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$

... from which we learn the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$



The **standard error** for $\hat{\beta}_0$ and $\hat{\beta}_1$ on a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$ are given by:

$$SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{N} + \frac{\bar{x}^2}{\sum_{i=1}^N (x_i - \bar{x})^2} \right], SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

, where $\sigma^2 = \text{Var}(\varepsilon)$



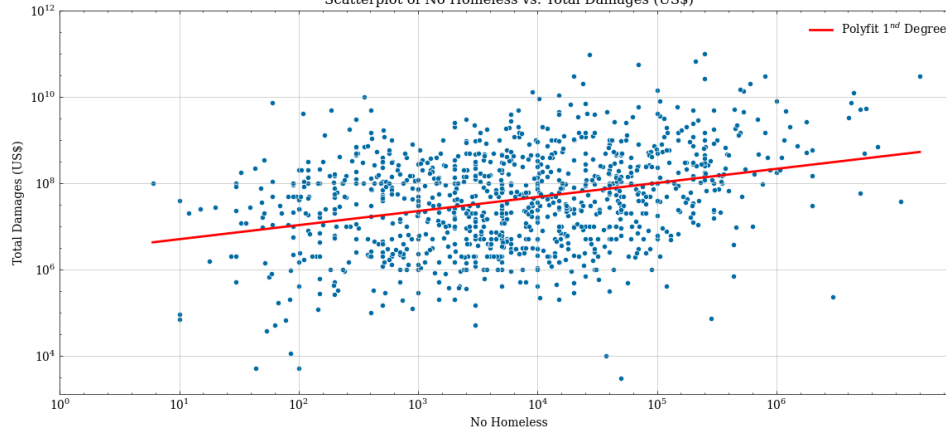
How do our observations $\{(x_i, y_i)\}_{i=1}^N$ shape the **standard error**?

Standard errors let us test whether a relationship is real, not noise.

→ **Standard errors** can also be used to perform hypothesis tests on the hypothesis coefficients.



Scatterplot of No Homeless vs. Total Damages (US\$)

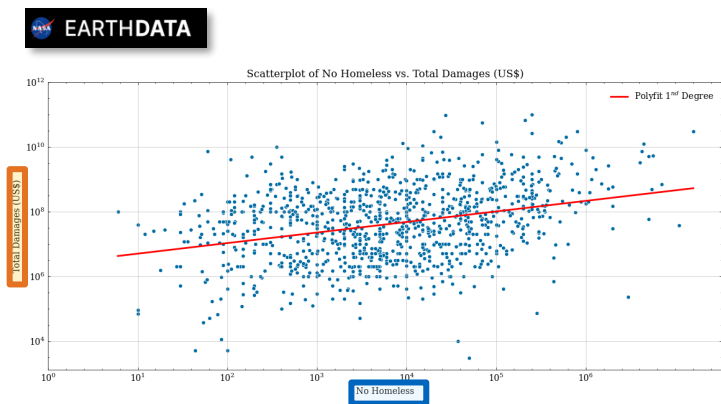


Hypothesis testing

H_0 : There is no relationship between X and Y versus the alternative hypothesis

H_1 : There is some relationship between X and Y

Our question, formalized: does the Homeless feature matter at all?



Recap that we want to learn a **model**:

$$\text{Total Damage (US\$)} \approx f(\text{Homeless, Total Death, Injured})$$

Consider the **linear model** and the learned coefficients $\hat{\beta}$ for our case at hand:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \times X_{\text{homeless}}$$

Mathematically the hypothesis test for our feature *homeless* corresponds to:

$$H_0: \hat{\beta}_1 = 0$$

$$H_1: \hat{\beta}_1 \neq 0$$

since if $\hat{\beta}_1 = 0$ then X_{homeless} is not associated with Y .

The t-statistic and p-value quantify the evidence against "no effect".

$$H_0: \hat{\beta}_1 = 0$$

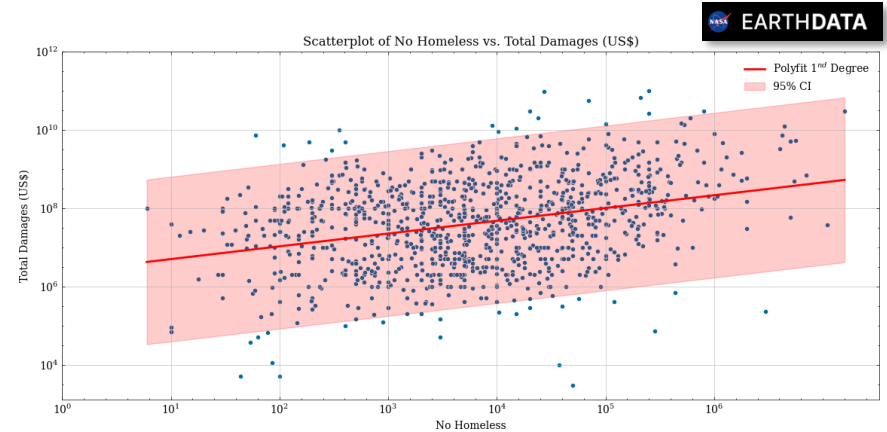
$$H_1: \hat{\beta}_1 \neq 0$$

To test the null hypothesis, we compute a **t-statistic**, given by:

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

We then compute the probability of observing any value equal to $|t|$ or larger. This probability is called **p-value**.

→ A small **p-value** indicates that it is unlikely to observe such a substantial association between the predictor and the response due to chance.



The **standard error** of an estimator $SE(\beta_1)$ reflects how it varies under repeated sampling.

These standard errors can be used to compute confidence intervals. A 95% **confidence interval** is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter:

$$[\hat{\beta}_1 - 2 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 2 \times SE(\hat{\beta}_1)]$$

Every decision rule risks a false alarm or a missed effect.

		Decision	
		fail to reject H_0	reject H_0
Truth	H_0 true	✓	Type 1 Error
	H_1 true	Type 2 Error	✓

We reject H_0 when the **p-value** is less than the **significance level** α . We often set α to be 0.01 or 0.05.

- This means that, for those cases where H_0 is actually true, we do not want to incorrectly reject it more than $100 \times \alpha$ % of those times.



So, for a significance level of $\alpha = 0.05$ how does our resulting model look?



EARTHDATA

OLS Regression Results

Dep. Variable:	Total Damages (US\$)	R-squared:	0.103			
Model:	OLS	Adj. R-squared:	0.102			
Method:	Least Squares	F-statistic:	121.8			
Date:	Wed, 17 Jun 2026	Prob (F-statistic):	6.92e-27			
Time:	14:06:22	Log-Likelihood:	-1590.3			
No. Observations:	1067	AIC:	3185.			
Df Residuals:	1065	BIC:	3195.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	6.3726	0.117	54.320	0.000	6.142	6.603
No Homeless	0.3266	0.030	11.035	0.000	0.269	0.385
Omnibus:	4.882	Durbin-Watson:	1.695			
Prob(Omnibus):	0.087	Jarque-Bera (JB):	4.813			
Skew:	-0.164	Prob(JB):	0.0901			
Kurtosis:	3.036	Cond. No.	15.0			

Mathematically the hypothesis test corresponds to:

$$H_0: \hat{\beta} = 0$$

$$H_1: \hat{\beta} \neq 0$$

- Simple linear regression
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RSS alone can't tell us how good the linear assumption is.

Based on a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$ we learned the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$

Such that it minimizes the **Residual sum of squares (RSS)**:

$$\min \sum_{i=1}^N (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$



How good is our **linear assumption** in describing our data?

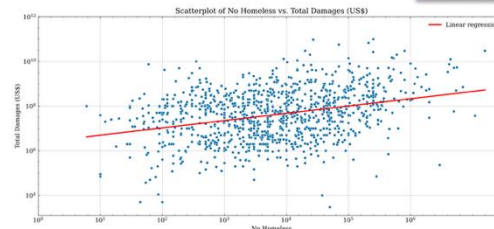
Linear Regression

Linear regression is a simple approach to supervised learning. We believe about the world that: $Y \approx f(X) + \varepsilon$.

In the case of linear regression, we assume that the dependence of Y on X is linear:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Annotations:
- β_0 : y-axis intercept
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True regression functions are never linear!

Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

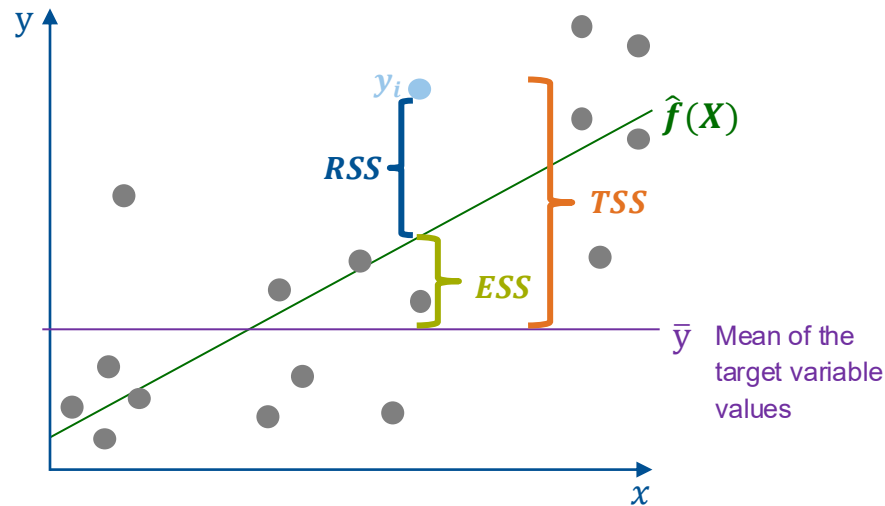
Total variation splits cleanly into explained and residual parts.

Based on a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$ we learned the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$

Such that it minimizes the **Residual sum of squares (RSS)**:

$$\min \sum_{i=1}^N (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$



How good is our
linearity assumption in
describing our data?

Total sum of squares (TSS)

$$\sum_{i=1}^N (y_i - \bar{y})^2$$

= **Residual sum of squares (RSS)**

$$= \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

+ **Explained sum of squares (ESS)**

$$+ \sum_{i=1}^N (\hat{y}_i - \bar{y})^2$$

R^2 is the share of variance our line actually explains.



How good is our
linearity assumption in
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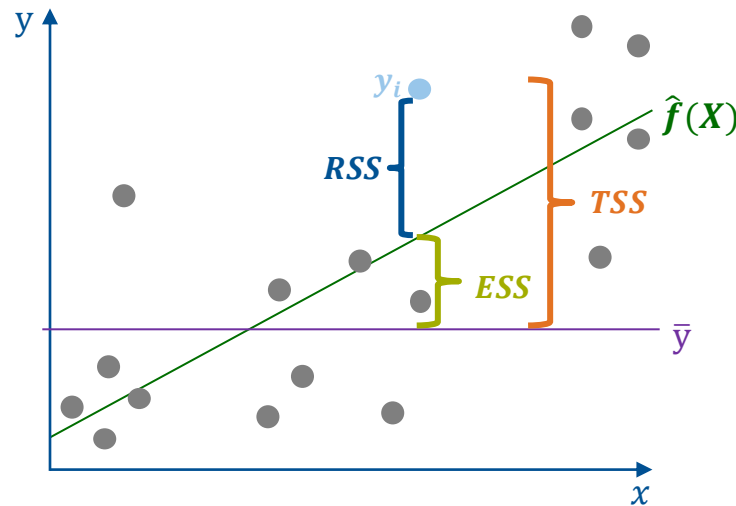
R^2 measures the proportion of the variation in y that is explained by the variation in x :

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} = \frac{ESS}{TSS}$$

$$R^2 \in [0, 1]$$

$R^2 = 1$: Perfect match between the line and the datapoints

$R^2 = 0$: no linear relationship between x and y



Total sum of squares (TSS)

$$\sum_{i=1}^N (y_i - \bar{y})^2$$

= Residual sum of squares (RSS)

$$= \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

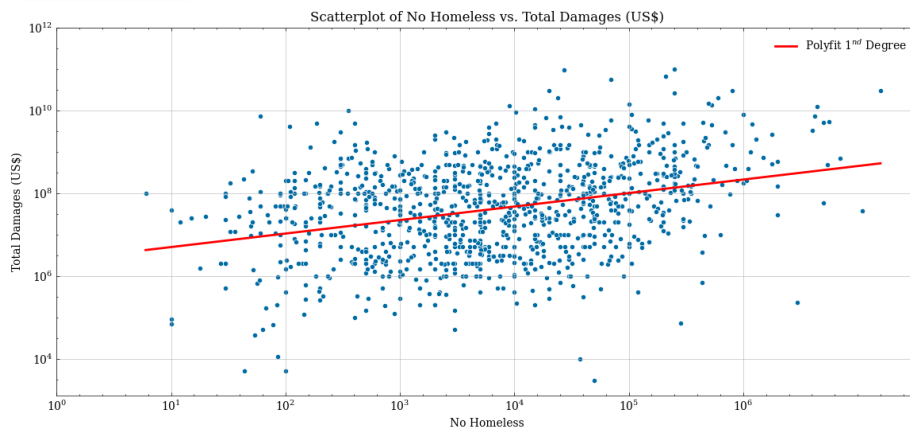
+ Explained sum of squares (ESS)

$$+ \sum_{i=1}^N (\hat{y}_i - \bar{y})^2$$

What do these coefficients actually say about disasters?

Remember our **model**:

$$\text{Total Damage (US\$)} \approx f(\text{Homeless})$$



OLS Regression Results

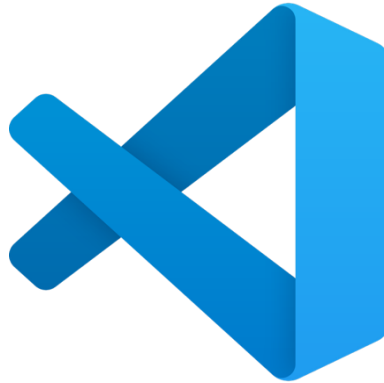
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How do we
interpret the
results?



- Simple linear regression
- **Multiple linear regression**
- Extensions to linear regression

With many predictors, the prediction is a weighted sum of them all.

What if we now consider: $X = (X_1, X_2, \dots, X_p)$?

Linear Regression

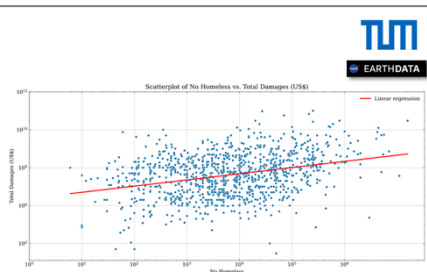
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$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Annotations: β_0 is the y-axis intercept, β_1 is the slope, and ε is the error term.

... but we can not observe this entity
... instead we have a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$
... from which we learn the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$



True regression functions are never linear!

Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

Then our belief about the world is:

$$Y = \beta_0 + \beta_1 \times X_1 + \beta_2 \times X_2 + \dots + \beta_p \times X_p + \varepsilon$$

And based on a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$, where x_i is now a p -dimensional vector, we learn the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \dots + \hat{\beta}_p X_p$$

RSS generalizes unchanged to p predictors.

What if we now consider: $X = (X_1, X_2, \dots, X_p)$?

Linear Regression

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

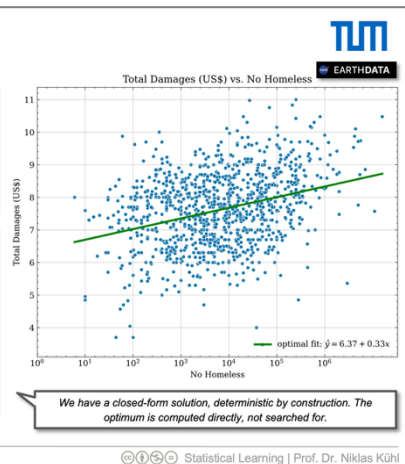
$$= \min \sum_{i=1}^N (y_i - \hat{\beta}_0 + \hat{\beta}_1 x_i)^2 = \min \sum_{i=1}^N \varepsilon_i^2$$

Residual sum of squares (RSS)

Using some calculus, one can show that the minimizers are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$



Residual sum of squares (RSS)

$$= \sum_{i=1}^N (y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i,1} - \hat{\beta}_2 X_{i,2} - \dots - \hat{\beta}_p X_{i,p})^2$$

Recap: Lecture 1 Basics

Input and target features can be formalized in matrixes.



Feature: an individual measurable property or characteristic of a phenomenon

We let $\mathbf{X}$ denote an $n \times p$ matrix whose (i, j) th element is $x_{i,j}$. That is,

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

Target: is the variable the model is trying to predict

We use y_i to denote i th observation of the variable on which we wish to make predictions. Hence, we write the set of all n observations in vector form as

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Rewriting RSS in matrix form.

What if we now consider: $X = (X_1, X_2, \dots, X_p)$?

Linear Regression

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$= \min \sum_{i=1}^N (y_i - \hat{\beta}_0 + \hat{\beta}_1 x_i)^2 = \min \sum_{i=1}^N \varepsilon_i^2$$

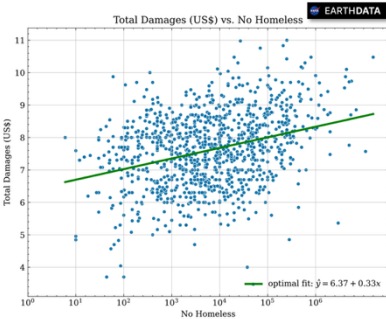
Residual sum of squares (RSS) 

Using some calculus, one can show that the minimizers are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

mean of all x mean of all y

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$



Total Damages (US\$) vs. No Homeless

optimal fit: $\hat{y} = 6.37 + 0.33x$

We have a closed-form solution, deterministic by construction. The optimum is computed directly, not searched for.

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Do we still have a closed form optimal solution?

Residual sum of squares (RSS)

$$= \sum_{i=1}^N (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i,1} - \hat{\beta}_2 x_{i,2} - \dots - \hat{\beta}_p x_{i,p})^2$$

We formulate in matrix notation $Y = X\beta + \varepsilon$ and get:

$$\sum_{i=1}^N (y_i - x_i \beta)^2$$

The matrix RSS expands into a term quadratic in β .

We formulate in matrix notation $Y = X\beta + \varepsilon$ and get **Residual sum of squares (RSS)**

$$= \sum_{i=1}^N (y_i - x_i\beta)^2$$

$$= (y - X\beta)'(y - X\beta)$$

$$= y'y - y'X\beta - \beta'X'y + \beta'X'X\beta$$

$$= y'y - 2\beta'X'y + \beta'X'X\beta$$

Following $\sum_i a_i^2 = a'a$

Expand the product, keep in mind, that transpose reverses the order of the factors $(AB)' = B'A'$

Identify that $y'X\beta$ is of size 1×1 and a scalar equals its own transpose

Differentiate with respect to β to identify the minimum $\frac{\partial RSS}{\partial \beta}$

We differentiate the RSS and set it to zero to identify the optimum.

$$\text{RSS} = y'y - 2\beta'X'y + \beta'X'X\beta$$

$$\frac{\partial \text{RSS}}{\partial \beta} = \frac{\partial (y'y - 2\beta'X'y + \beta'X'X\beta)}{\partial \beta}$$

$$= \frac{\partial (y'y)}{\partial \beta} - \frac{\partial (2\beta'X'y)}{\partial \beta} + \frac{\partial (\beta'X'X\beta)}{\partial \beta}$$

$$= -2X'y + 2X'X\beta$$

$$\Rightarrow X'X\beta = X'y$$

$$\beta = (X'X)^{-1}X'y$$

Differentiate with respect to β to identify the minimum $\frac{\partial \text{RSS}}{\partial \beta}$

Independent of $\beta \rightarrow$ drops

By calculus we know $\frac{\partial (a'b)}{\partial b} = a$

By calculus we know $\frac{\partial (b'Ab)}{\partial b} = 2Ab$ for symmetrical A

Set gradient to zero $\frac{\partial \text{RSS}}{\partial \beta} = 0$

If $X'X$ is invertible, multiply both sides on the left by $(X'X)^{-1}$

Again: We have a closed-form solution, deterministic by construction. The optimum is computed directly, not searched for.

Second derivative confirms a minimum.

$$\beta = (X'X)^{-1}X'y$$

*Again: We have a closed-form solution, deterministic by construction.
The optimum is computed directly, not searched for.*



Is it really the min
RSS, not the max?

The second derivative (Hessian) is $2X'X$, which is positive semidefinite for any X , so the stationary point is the minimum, not a maximum or saddle.



How is this solution
different to the $p=1$ case?

*It's the same! Set $X=[1 \ x]$ into $(X'X)^{-1}X'y$ you get
 $\hat{\beta}_1 = \frac{\text{Cov}(X,Y)}{\text{Var}(X)}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$*



How is this solution different to the $p=1$ case?

It's the same! Set $X=[1 \ x]$ into $(X'X)^{-1}X'y$ you get $\hat{\beta}_1 = \frac{\text{Cov}(X,Y)}{\text{Var}(X)}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

$$X'X = \begin{pmatrix} N & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \quad X'y = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}$$

Decompose the two terms

$$(X'X)^{-1} = \frac{1}{D} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & N \end{pmatrix}, \quad D = N\sum x_i^2 - (\sum x_i)^2$$

Invert the 2×2 matrix (D = determinant)

$$\hat{\beta} = \frac{1}{D} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & N \end{pmatrix} \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \frac{1}{D} \begin{pmatrix} \sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i \\ N \sum x_i y_i - \sum x_i \sum y_i \end{pmatrix}$$

Multiply out and read off the two rows

$$\hat{\beta}_1 = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

First normal equation: $N\hat{\beta}_0 + (\sum x_i)\hat{\beta}_1 = \sum y_i$
 $\Rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Beyond the scope of this course.

One formula solves least squares for any number of predictors.

What if we now consider: $X = (X_1, X_2, \dots, X_p)$?

Linear Regression

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$= \min \sum_{i=1}^N (y_i - \hat{\beta}_0 + \hat{\beta}_1 x_i)^2 = \min \sum_{i=1}^N \varepsilon_i^2$$

Residual sum of squares (RSS)

Using some calculus, one can show that the minimizers are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

mean of all x mean of all y

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

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Residual sum of squares (RSS)

$$= \sum_{i=1}^N (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i,1} - \hat{\beta}_2 x_{i,2} - \dots - \hat{\beta}_p x_{i,p})^2$$

We formulate in matrix notation $Y = X\beta + \varepsilon$ and get the solution by:

$$\beta = (X'X)^{-1}X'y$$

Use t-tests for single coefficients, the F-test for the model as a whole.

Assess whether explanatory variables have significant effects

Individual significance (t-test)

- Assumption: sampled coefficients normally distributed with mean true coefficient
- Two-sided t-test for significant difference from zero

$$t = \frac{\hat{\beta} - 0}{SE(\hat{\beta})}$$

Find out whether explanatory variable influences dependent variable

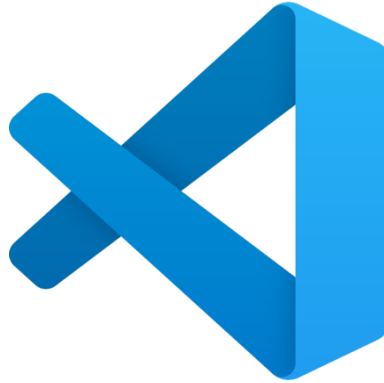
Joint significance (F-test)

- Test whether the model explains more than the intercept (=mean) alone
- Compared explanatory power added by variables to remaining unexplained variance

$$F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)}$$

Find out whether regressors jointly influence dependent variable

*P is the number of regressors including the intercept
N is the number of instances we derive the coefficients from*



- Simple linear regression
- Multiple linear regression
- Extensions to linear regression
 - Dummy variables
 - Modeling feature interaction
 - Coefficient shrinkage methods

We can include qualitative predictors via dummy variables.

Not every predictor is numeric — e.g. we can differentiate events by **Disaster Type** being a wildfire or not. A two-level factor becomes a single 0/1 indicator.

1. Encode the factor as an indicator

$$x_{i,1} = \begin{cases} 1 & \text{if disaster } i \text{ is a Wildfire} \\ 0 & \text{if disaster } i \text{ is any other Disaster Type} \end{cases}$$

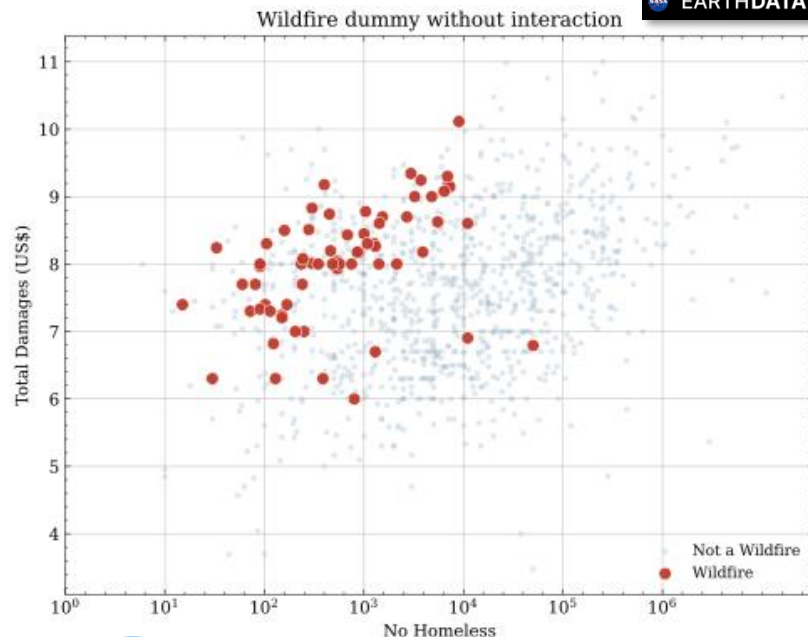
Lets $x_{i,2}$ be the No. Homeless

2. Put it in the regression

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \beta_2 x_{i,2} + \varepsilon_i & \text{Wildfire} \\ \beta_0 + \beta_2 x_{i,2} + \varepsilon_i & \text{any other} \end{cases}$$



How do we expect this regression to look?



What if we believe β_2 to be different for Wildfires?

- Simple linear regression
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And we can model interactions along qualitative and quantitative predictors.

Add a product term (Wildfire $\times$ log Homeless)

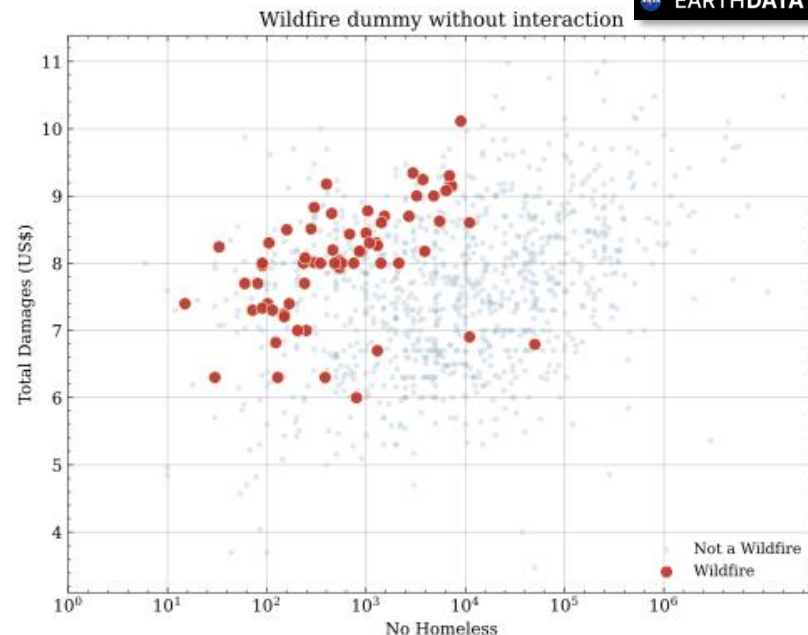
$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,1} x_{i,2} + \varepsilon_i$$

Split by the two cases

Wildfire ($x_{i,1} = 1$): $y_i = (\beta_0 + \beta_1) + (\beta_2 + \beta_3) x_{i,2} + \varepsilon_i$

Otherwise ($x_{i,1} = 0$): $y_i = \beta_0 + \beta_2 x_{i,2} + \varepsilon_i$

Now the slope on **Homeless** is β_2 for other disasters but $\beta_2 + \beta_3$ for wildfires. β_3 is the difference in slope — non-zero β_3 relaxes the additive assumption, so the lines are no longer parallel.



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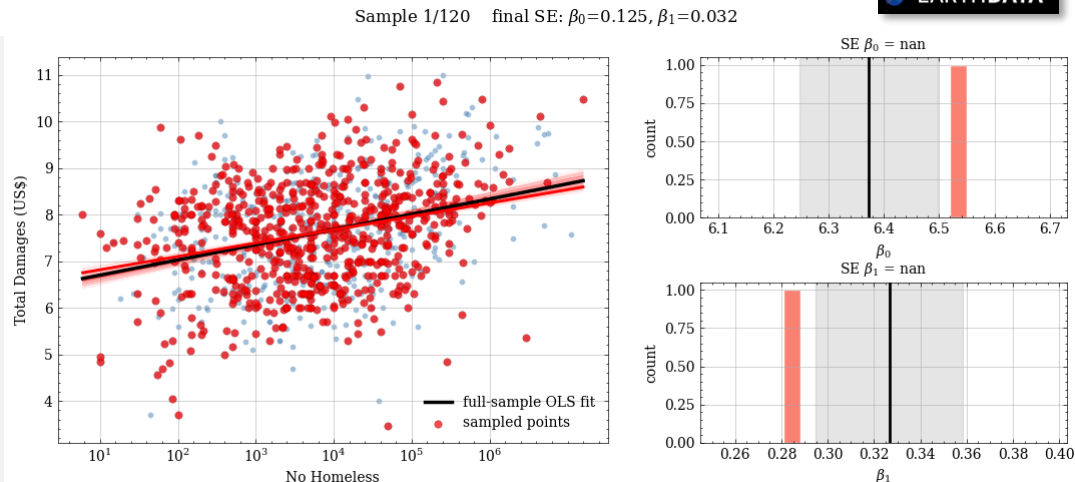
To generalize, we deliberately shrink coefficients toward zero.

Recall our objective of minimizing
Residual sum of squares (RSS):

$$\min \sum_{i=1}^N \left(y_i - \hat{\beta}_0 - \sum_{j=1}^p \hat{\beta}_p x_{i,j} \right)^2$$

And its closed form solution:

$$\beta = (X'X)^{-1}X'y$$



This is great for our finite set of observations $\{(x_i, y_i)\}_{i=1}^N$. But eventually we want to generalize to unseen data.

*We know that the derived coefficients are subject to variance in the data. And we can account for it by nudging our model towards smaller coefficients by **shrinkage methods**.*

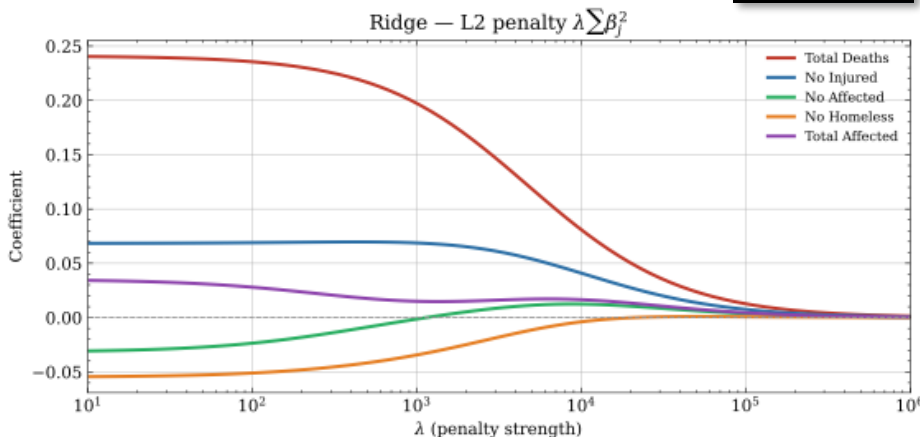
Ridge adds an L2 penalty that shrinks all coefficients smoothly.

We know that the derived coefficients are subject to variance in the data. And we can account for it by nudging our model towards smaller coefficients by **shrinkage methods**.

We adapt our minimizing objective:

$$\min \sum_{i=1}^N \left(y_i - \hat{\beta}_0 - \sum_{j=1}^p \hat{\beta}_p x_{i,j} \right)^2 + \lambda \sum_{j=1}^p \hat{\beta}_p^2$$

Linear regression with this L-2 regularization term is called **Ridge regression**.

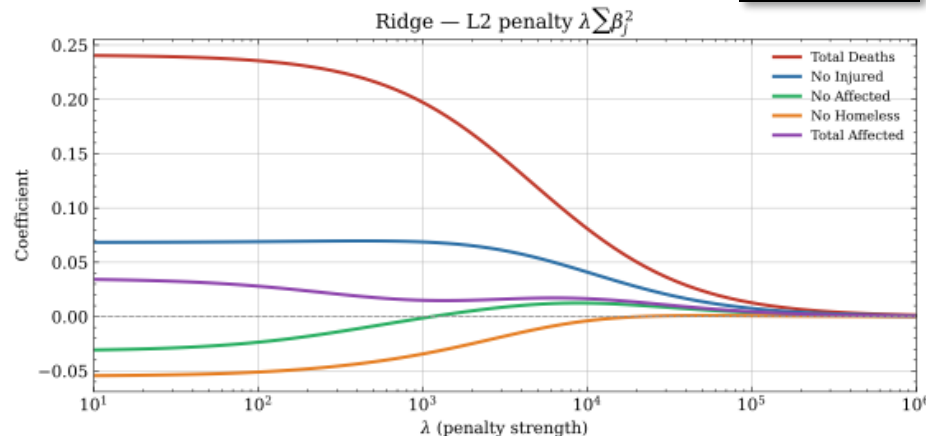


λ is a tuning parameter. Selecting a good value for λ is critical. As λ increases, the flexibility of the ridge regression fit decreases, leading to decreased variance but increased bias.

Few things to keep in mind when applying Ridge regression.

Linear regression with this L-2 regularization term is called **Ridge regression**.

$$\min RSS + \lambda \sum_{j=1}^p \hat{\beta}_j^2$$



?

Why $j=1$ not $j=0$?

We do not want to shrink the intercept, which is simply a measure of the mean value of the response variable.

?

What becomes critical about our predictors when we apply shrinkage?

Scaling the predictors. Contrast with OLS, which is scale-equivariant: rescale a predictor and its coefficient rescales inversely, leaving predictions, RSS, and significance untouched.

Lasso's L1 penalty can zero coefficients out, doing feature selection.

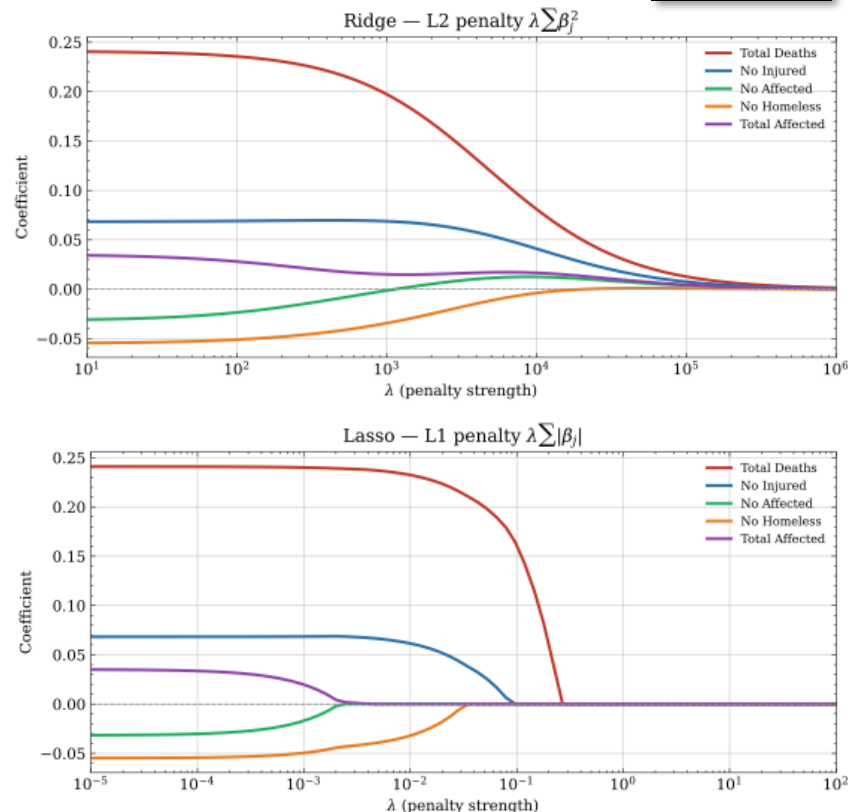
What if we not only want to shrink coefficients but force our model to select a subset of predictors?

Linear regression with this L-1 regularization term is called **Lasso regression**.

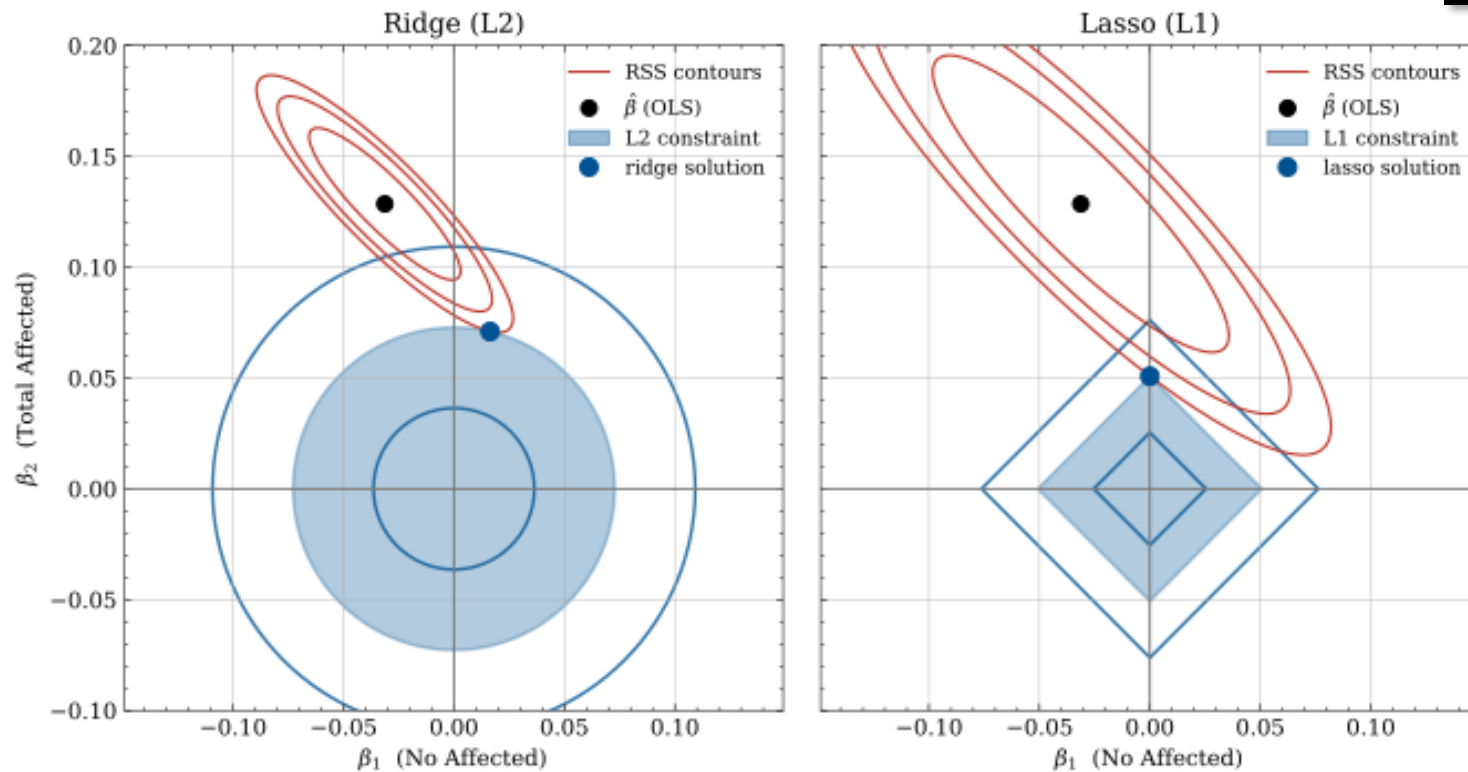
$$\min RSS + \lambda \sum_{j=1}^p |\hat{\beta}_j|$$



Why does **L-1 regularization** (**Lasso**) lead to feature selection?



The L1 diamond's corners are why lasso drives coefficients to exactly zero.



- Simple linear regression
 - Estimating the Coefficients
 - Assessing the Accuracy of the Coefficient Estimates
 - Assessing the Accuracy of the Model
- Multiple linear regression
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What is still **unclear** from this session?